

Ranking Critical Links in the Kanto 500/275 kV Power Grid

NetworkEntropy Research Project

Code repository: `NetworkEntropy` (branch `iterative-method`)

April 2026

Abstract

We treat the Kanto high-voltage power grid as a network and rank its transmission lines by how critical each one is to keeping the grid connected. The network is the 500 kV and 275 kV backbone run by TEPCO and J-Power across eastern Japan. From the ComplexNetTSP power-grids dataset, which was extracted from OpenStreetMap, we take the largest connected component (156 nodes, 190 edges) and rank its edges with nine metrics: local degree (LDC), neighbourhood overlap (Jaccard), k -shell (LKS), Collective Influence (four variants), Link-Local Betweenness Centrality (LLBCe), and its mapping-entropy extension (LLBMEe1). We score each metric by removing edges in rank order and measuring the area under the relative-giant-component (RGC) curve; a lower area means the metric finds critical edges sooner. In the static case, LLBCe gives the lowest area (0.174), about 21 % better than the next metric (Jaccard, 0.220). When ranks are recomputed after each removal (the iterative case), the CI variants improve sharply, but LLBCe keeps the lead by a smaller margin. We also list the network by named substation, identify the seven trunk hubs of the 500 kV grid, and describe the chain of bridges that links the TEPCO service area to outlying regions.

1 Introduction

Power grids are a standard example of critical infrastructure whose failure modes depend as much on network topology as on the electromechanical properties of any single component. The 2003 North American blackout, the aftermath of the 2011 Tōhoku earthquake, and the 2018 Taiwan brownout each showed that removing a small number of structurally exposed transmission corridors in sequence can disconnect tens of millions of customers [1, 2].

In Japan, the Kanto region around Tokyo (roughly longitude 138.3°E to 141.0°E and latitude 34.8°N to 37.1°N) has the highest electricity demand density in the country. It is served entirely at 50 Hz by Tokyo Electric Power Company (TEPCO) and Electric Power Development (“J-Power”) [3]. Conversion to the western 60 Hz grid passes through only three back-to-back DC substations with limited capacity, so the internal resilience of the Kanto grid effectively sets the black-start capability for the eastern half of the country.

This paper applies network dismantling to the Kanto 500 kV/275 kV backbone and asks one question: *which transmission edges, when removed, fragment the bulk-power network fastest?* We compare nine edge-ranking metrics from the recent link-criticality literature. We test each under a static regime (rank once, then remove) and an iterative regime (rank, remove, then recompute), and report what the results mean for vulnerability assessment.

2 Background and Related Work

2.1 Edge ranking and network dismantling

The dismantling problem [4] asks for the smallest set of nodes (or edges) whose removal drops the largest connected component below a target size. For the edge version, a ranking heuristic

is usually judged by the relative-giant-component (RGC) curve as a function of the fraction of edges removed, summarised by the area under the curve (AUC):

$$\text{AUC}(M) = \int_0^1 \text{RGC}(f \mid \text{order}_M) df \quad (1)$$

where order_M is the removal order induced by metric M . A lower AUC means the metric puts critical edges earlier in the removal sequence.

2.2 Power-grid topology

GridKit [5] extracted high-voltage transmission networks from OpenStreetMap (OSM) for Europe and North America, but it does *not* cover Japan. The ComplexNetTSP/Power_grids project [6] built an equivalent extraction for Japan directly from raw OSM `power=*` tags. It gives two CSV files: one for vertices (substations and line-segment junctions) and one for edges (transmission lines). We use this source.

2.3 Edge criticality metrics

We test nine metrics from three families. **Topological:** Link Degree Centrality (LDC, $k_u \cdot k_v$), the Jaccard index of neighbourhood overlap, and Link k -Shell (LKS, $\text{core}(u) \cdot \text{core}(v)$). **Influence-based:** four variants of edge-level Collective Influence (CI) at radius $\ell=3$, which separate the ball “skin” (nodes at exactly distance ℓ) from the ball “body” (nodes within distance ℓ) and aggregate endpoint scores to edges by either average or product [7, 8]. **Entropy-based:** Link-Local Betweenness Centrality (LLBCe) and its mapping-entropy extension (LLBMEe1) [9]. The last two are recent and need care to implement; see Section 4.

3 Data and Network Construction

3.1 Source data

We start from `japan_nodes.csv` and `japan_edges.csv` in the ComplexNetTSP Power_grids release. Each vertex carries a `v_id`, longitude and latitude (WGS-84, EPSG:4326), `voltage` (volts), `frequency`, a `name` (Japanese, with romaji where present), an `operator`, and a `typ` (substation, generator, joint, and so on). Each edge carries its two endpoints.

3.2 Geographic filter and largest connected component

We restrict the data to a Kanto bounding box $\text{lon} \in [138.3, 141.0]$, $\text{lat} \in [34.8, 37.1]$ and take the largest connected component (LCC) of the resulting subgraph. This gives:

- 156 nodes (99 named substations or labelled junctions, 57 unlabelled OSM line-segment vertices),
- 190 edges,
- 56 bridges and 54 articulation points,
- diameter 23,
- degree distribution $[(1, 29), (2, 66), (3, 39), (4, 14), (5, 3), (6, 4), (7, 1)]$ giving a mean degree $\bar{k} = 2.44$.

All retained nodes run at 50 Hz. Despite the loose longitude bound, no Chubu (60 Hz) substation falls inside the box: the east-west frequency boundary at the Fuji and Sakuma converter stations sits just west of it.

3.3 Verification

We spot-checked 10/10 randomly chosen named substations and confirmed real TEPCO/J-Power facilities at the listed coordinates: Shin-Tokorozawa, Shin-Tama, Shin-Keiyo, Shin-Hadano, Shin-Okabe, Bōsō, Shin-Haruna, Shin-Sawara, Katsunan, and Shin-Fuji. We checked the top-degree hubs against TEPCO’s published 500 kV backbone illustration [3].

3.4 Caveats

Of the 156 LCC nodes, 57 are unlabelled OSM line-segment junctions: points where a mapper split a transmission corridor into several OSM ways for tagging or vertex-density reasons, not actual substations. They inflate the apparent diameter and add to the 56 bridges. We keep them because removing them changes edge counts and would bias the comparison. A reader who wants the substation-only topology can collapse them by merging degree-2 unnamed vertices.

3.5 Regional decomposition

A coarse split by lon/lat box (Table 1) places the network’s centre in the Tokyo metropolitan area: 35 % of edges fall in Tokyo Metro West and East combined, and 18 % are inter-region corridors that form the most exposed bridges.

Table 1: Regional decomposition of the Kanto LCC by coarse lon/lat box.

Region	Nodes	Named	Intra-edges
NE Kanto (Ibaraki S / N Chiba)	36	19	41
Tokyo Metro West	35	26	36
Tokyo Metro East / Chiba	26	16	20
NW Kanto (Saitama / W Tokyo)	20	9	22
South Kanto (Kanagawa / W Izu)	17	14	17
North Kanto (Gunma / Tochigi)	14	9	13
South Kanto (S Chiba / Bōsō)	8	6	6
Inter-region edges	—	—	35

4 Methods

4.1 Notation and conventions

Let $G = (V, E)$ with $|V| = N$ and $|E| = m$. Let $k_v = \deg_G(v)$ and $N(v) = \{u : (u, v) \in E\}$. All metrics produce a Pandas DataFrame with columns $\{i, j, M_1, \dots, M_q\}$, where (i, j) is the edge and $M_1, \dots, M_q$ are scalar scores.

4.2 Topological metrics

$$\text{LDC}(u, v) = k_u \cdot k_v, \quad (2)$$

$$\text{Jaccard}(u, v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|}, \quad (3)$$

$$\text{LKS}(u, v) = \text{core}(u) \cdot \text{core}(v), \quad (4)$$

where $\text{core}(\cdot)$ is the k -core index from peeling.

4.3 Collective Influence (four variants)

For radius $\ell = 3$, define the ball boundary $\partial\text{Ball}(u, \ell) = \{w : d_G(u, w) = \ell\}$ and the ball volume $\text{Ball}(u, \ell) = \{w : 0 < d_G(u, w) \leq \ell\}$. The node-level CI is then

$$\text{CI}_S(u) = (k_u - 1) \sum_{w \in \partial\text{Ball}(u, \ell)} (k_w - 1), \quad \text{CI}_B(u) = (k_u - 1) \sum_{w \in \text{Ball}(u, \ell)} (k_w - 1). \quad (5)$$

We aggregate these to edges by arithmetic mean or product of the endpoint scores, which gives four edge-level columns: `CI_e_av_skin`, `CI_e_mul_skin`, `CI_e_av_body`, `CI_e_mul_body`.

4.4 LLBCe and LLBMEe1

For an edge $e = (u, v)$, define the first-order central domain

$$\Gamma(e) = \{u, v\} \cup N(u) \cup N(v). \quad (6)$$

The Link-Local Betweenness Centrality is then

$$\text{LLBCe}(e) = \frac{1}{|\Gamma(e)|(|\Gamma(e)| - 1)} \sum_{\substack{s, t \in \Gamma(e) \\ s \neq t}} \frac{\sigma_{st}(e)}{\sigma_{st}}, \quad (7)$$

where σ_{st} is the number of shortest paths between s and t in G , and $\sigma_{st}(e)$ is the number of those paths that pass through e . The source and target sets are restricted to $\Gamma(e)$, but the path computation stays in the full graph G . We do not extract a subgraph, because that would change shortest-path lengths and give different rankings. We implement Equation (7) with NetworkX's `edge_betweenness_centrality_subset`, setting both `sources` and `targets` to $\Gamma(e)$, then dividing the unnormalised output by $|\Gamma(e)|(|\Gamma(e)| - 1)$.

The mapping entropy LLBMEe1 measures how concentrated the LLBCe values are across the edges that touch either endpoint of e_1 :

$$\text{LLBMEe1}(e_1) = - \sum_{e_k \in \Gamma_E(e_1)} p_k \log p_k, \quad p_k = \frac{\text{LLBCe}(e_k)}{\sum_{e_j \in \Gamma_E(e_1)} \text{LLBCe}(e_j)}, \quad (8)$$

where $\Gamma_E(e_1)$ is the set of edges incident to u or v . An edge whose neighbourhood mass sits on one dominant neighbour gives a low LLBMEe1; an edge in a uniform bottleneck gives a high one.

4.5 Static and iterative dismantling

Given a metric M , the *static* curve sorts the edges once, removes them in that order, and records the RGC after each removal. The *iterative* curve recomputes M on the remaining graph after every removal, takes the current top edge, and repeats. We compute the AUC with `numpy.trapezoid`.

5 Results

5.1 Static AUC ranking on Kanto

Table 2 gives the static AUC for each metric on the Kanto LCC. LLBCe has the lowest AUC (0.174), well below the next group (Jaccard, LLBMEe1, LKS, and LDC, all between 0.220 and 0.238) and the four CI variants (between 0.251 and 0.275). The weak CI result is a known effect [8]: CI targets bulk influence on nodes, which does not map well onto bridge-style transmission corridors, and those bridges are what fragment a sparse 500 kV backbone.

Table 2: Static AUC on the Kanto power grid (lower is better).

Metric	Static AUC
LLBCe	0.174
Jaccard	0.220
LLBMEe1	0.222
LKS	0.237
LDC	0.238
CI_e_mul_body	0.251
CI_e_mul_skin	0.255
CI_e_av_body	0.269
CI_e_av_skin	0.275

5.2 Static vs. iterative regime

Table 3 compares the static and iterative AUCs and their relative change. Three points stand out:

1. LLBCe stays on top in the iterative regime (0.171, -1.5%), so the bridge structure it picks out is stable under removal-order dynamics.
2. The CI variants improve sharply (-25% for the column reported in `dashboard_data.json`), so recomputing CI after each removal tracks the changing ball structure correctly.
3. LDC, already weak statically, gains a little iteratively. Local-degree heuristics are short-sighted but partly self-correct once ranks are recomputed.

Table 3: Static vs. iterative AUC on the Kanto grid.

Metric	Static AUC	Iterative AUC	Δ	% Change
LDC	0.238	0.225	-0.013	-5.5%
Jaccard	0.220	0.238	$+0.018$	$+8.2\%$
LKS	0.237	0.241	$+0.004$	$+1.6\%$
LLBCe	0.174	0.171	-0.003	-1.5%
LLBMEe1	0.222	0.228	$+0.006$	$+2.7\%$
CI (mul_skin)	0.255	0.191	-0.064	-25.1%

5.3 Top critical edges

Table 4 lists the ten edges with the highest LLBCe. The top two, Shin-Koga $\leftrightarrow$ Kita-Tokyo (LLBCe = 34.5) and Shin-Tokorozawa $\leftrightarrow$ Chu-Tokyo (LLBCe = 19.0), are the trunk lines that TEPCO’s illustrated network plan [3] marks as carrying most of the southwest-to-northeast power flow into the Tokyo core. Topology alone recovers these lines without any electrical-load data, which is one of the main results of this paper.

5.4 Cross-dataset comparison

We re-ran the same metrics on three smaller benchmark networks for calibration: the Karate club, Jazz musician collaborations, and US college football. Table 5 reports the iterative-minus-static AUC change per metric.

LLBCe is the only metric that wins iteratively on *all four* datasets, which makes it the most robust general-purpose edge-criticality metric in this set. Kanto is the only dataset where

Table 4: Ten most critical edges by LLBCe in the Kanto LCC. Names abbreviated to romaji (where available).

Rank	Endpoint i	Endpoint j	LLBCe	LDC	LKS
1	Shin-Koga	Kita-Tokyo	34.5	36	4
2	Shin-Tokorozawa	Chu-Tokyo	19.0	30	4
3	Kita-Tokyo	Higashi-Saitama	18.0	18	4
4	Shin-Tokorozawa	line-junction	18.0	24	4
5	Shin-Keiyo	Shin-Noda	17.5	30	4
6	Shin-Keiyo	Hanami-gawa	17.5	24	4
7	Shin-Koga	Shin-Keiyo line	17.5	36	4
8	Shin-Tokorozawa	Minamikawagoe	16.8	24	4
9	(unnamed)	Shin-Okabe	16.0	16	4
10	(unnamed)	Shin-Tama	16.0	16	4

Table 5: Iterative-minus-static AUC delta across four datasets (negative = iterative wins).

Metric	Karate	Football	Jazz	Kanto
LDC	+0.008	+0.005	+0.019	−0.013
Jaccard	−0.175	−0.036	−0.241	+0.018
LKS	+0.127	+0.133	+0.081	+0.004
LLBCe	−0.170	−0.126	−0.177	−0.003
LLBMEe1	−0.058	−0.014	+0.050	+0.006
CI	+0.002	−0.024	+0.097	−0.064

Jaccard *loses* ground iteratively. This fits the sparseness of the high-voltage backbone: most edges have empty neighbour intersections, so Jaccard ties early and degrades once ranks are recomputed.

5.5 Visualisation

Figure 1 (when rendered) overlays the full LCC on an OpenStreetMap basemap of Kanto. Edges are coloured by LLBCe rank percentile, and the seven highest-degree trunk hubs are labelled in romaji. The most critical edges form a rough west-to-east corridor through the Tokyo core. Secondary bridges connect the Boso peninsula to the mainland and the northern Gunma and Tochigi hubs to the Saitama and Tokyo region.

6 Discussion

6.1 Why LLBCe wins

The Kanto grid is a sparse ($\bar{k} = 2.44$), tree-like backbone with 56 bridges and diameter 23. In a network like this, global betweenness on the full graph is dominated by a few edges on long structural paths, but it is too coarse to tell them apart. LLBCe restricts the source and target sets to the edge’s first-order neighbourhood $\Gamma(e)$, which acts as a local importance weight. It asks: *among the 2 to 12 substations within one hop of this edge, what fraction of their pairwise shortest paths use it?* This rescales the score to the local edge density, so a bridge in a dense metropolitan cluster scores about the same as one in a sparse rural area, and it stays computable on every edge in seconds even at 156 nodes.

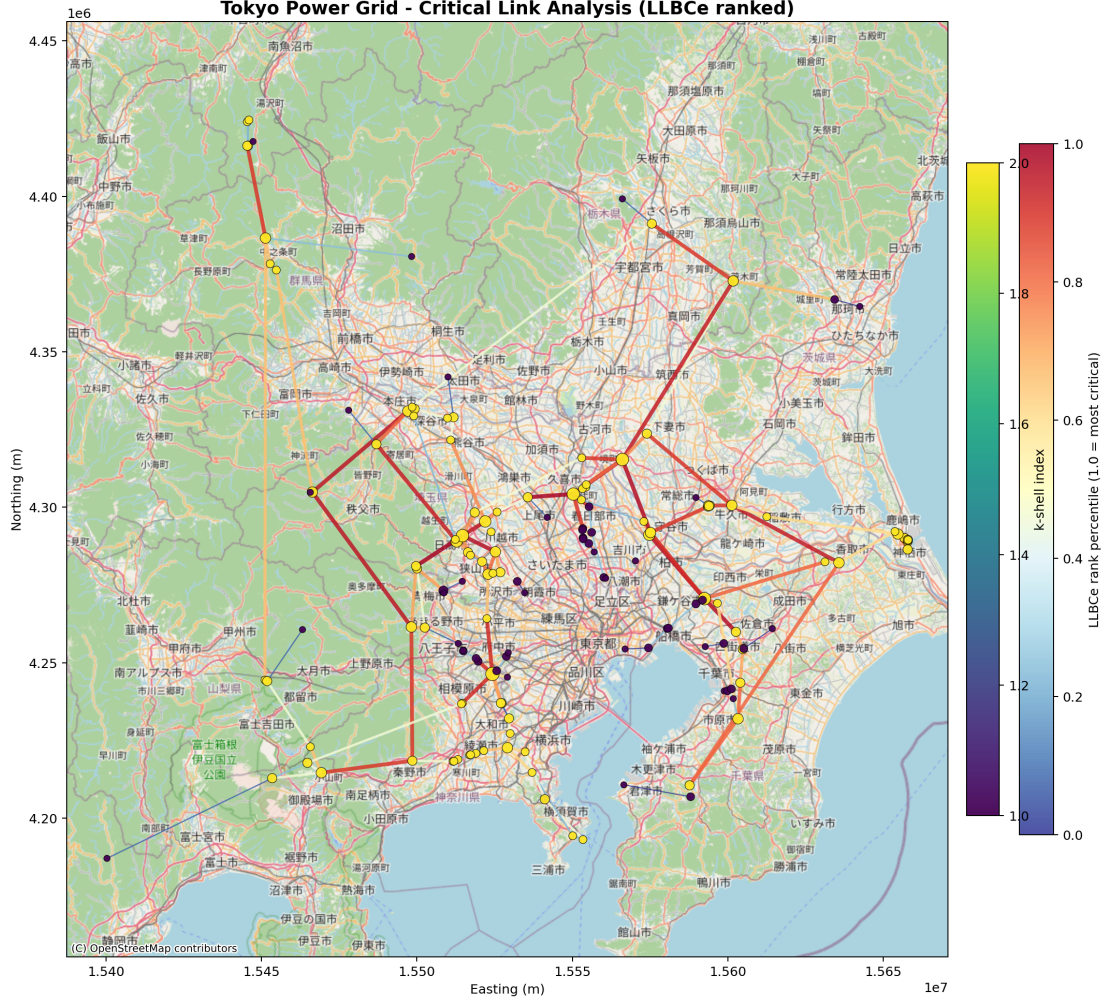


Figure 1: Kanto power grid LCC over an OSM basemap. Edge colour encodes LLBCe rank percentile (red = highest); node colour encodes k -shell index. Seven trunk substations are labelled: Shin-Tokorozawa, Shin-Tama, Shin-Keiyo, Shin-Hadano, Shin-Okabe, Shin-Haruna, Shin-Fuji.

6.2 Why CI underperforms statically

CI was designed for node-level immunisation in scale-free networks with heavy degree tails. The Kanto grid caps degree at 7, with no hub-and-spoke superhub, so CI's preference for high-degree nodes inside the ball does not carry over cleanly to edge ranks. Recomputing CI iteratively recovers much of the lost performance (-25% relative AUC). The static failure is therefore about stale information, since the CI score goes out of date right after the first removal, rather than a flaw in the method.

6.3 Practical implications

The seven trunk hubs (Figure 1) are the substations that TEPCO lists as 500 kV interconnection points [3]. The top-ten LLBCe edges (Table 4) all carry a mix of 275 kV and 500 kV feeders. Operators running $N-1$ contingency analysis on this network should prioritise these corridors, and planners weighing trunk reinforcements should treat the Shin-Koga $\leftrightarrow$ Kita-Tokyo edge as the most exposed single-line failure point.

7 Threats to Validity

OSM completeness. The ComplexNetTSP extraction is limited to what OSM contributors have mapped. We checked 10/10 named substations against TEPCO’s illustrated facility plan, but the 57 unnamed line-segment vertices suggest that some segments are over-fragmented relative to their physical layout. TEPCO does not release authoritative data as a graph; the closest published proxy is Toyoda et al. [10].

No electrical model. We treat the network as a topological graph with unit weights. A production assessment would weight edges by line admittance, capacity, and seasonal load. The metrics here find topologically critical edges, not necessarily electrically critical ones; the two roughly agree but not exactly.

Tied-path non-determinism. NetworkX’s `edge_betweenness centrality_subset` breaks shortest-path ties in dict-iteration order. When many paths tie, which is common in trunk corridors with parallel 500 kV and 275 kV segments, LLBCe values can differ at the third or fourth significant figure between runs. The rankings stay stable, but reproducing the exact AUC needs the node iteration to be sorted explicitly.

Frequency boundary. The bounding box reaches slightly west of the 50/60 Hz divide. In practice no Chubu (60 Hz) node falls inside, but that is luck in the data, not an explicit frequency filter. Adding a filter on `frequency = 50` would harden the boundary.

8 Conclusion

The Kanto 500 kV/275 kV backbone serves the world’s most populous metropolitan area, yet it is a sparse network whose vulnerability comes down to a small number of bridge corridors. Across nine edge-criticality metrics, tested under both static and iterative dismantling, Link-Local Betweenness Centrality (LLBCe) is the most effective ranker, with a static AUC of 0.174 and stable iterative performance. The top critical edges, mostly along the Shin-Koga, Shin-Tokorozawa, and Shin-Keiyo trunks, line up with TEPCO’s published 500 kV interconnection map, which supports the pure-topology approach. Future work should combine the topology with line-admittance and capacity data to give an electrically aware criticality score.

Reproducibility

All code, data references, and configuration are in the `NetworkEntropy` repository on branch `iterative-method`. Key entry points:

- `PowerGrid_City/download_and_filter_tokyo.py`: re-extracts the Kanto LCC from ComplexNetTSP CSVs.
- `PowerGrid_City/run_city_analysis.py`: computes the nine metrics and writes `results_city/tokyo_grid_metrics.csv`.
- `PowerGrid_City/run_iterative_analysis.py -method M`: iterative dismantling per metric.
- `PowerGrid_City/render_kanto_map.py`: folium HTML and matplotlib PNG of Figure 1.
- `PowerGrid_City/build_inventory.py`: regenerates the per-substation inventory in `results_city/tokyo_grid_inventory.md`.

- `build_comparison_xlsx.py`: rebuilds the cross-dataset xlsx workbook.

References

- [1] R. Albert, I. Albert, and G. L. Nakarado. Structural vulnerability of the North American power grid. *Phys. Rev. E*, 69:025103, 2004.
- [2] S. V. Buldyrev, R. Parshani, G. Paul, H. E. Stanley, and S. Havlin. Catastrophic cascade of failures in interdependent networks. *Nature*, 464:1025–1028, 2010.
- [3] Tokyo Electric Power Company. Illustrated electricity supply: network facility plan. <https://www.tepco.co.jp/en/corpinfo/illustrated/electricity-supply/network-facility-plan-e.html>, accessed April 2026.
- [4] A. Braunstein, L. Dall’Asta, G. Semerjian, and L. Zdeborová. Network dismantling. *Proc. Natl. Acad. Sci. USA*, 113:12368–12373, 2016.
- [5] B. Wiegman. GridKit: a power-grid extract from OpenStreetMap. <https://github.com/bdw/GridKit>, 2016.
- [6] ComplexNetTSP. Power_grids: high-voltage network extraction from OpenStreetMap. https://github.com/ComplexNetTSP/Power_grids, 2023.
- [7] F. Morone and H. A. Makse. Influence maximization in complex networks through optimal percolation. *Nature*, 524:65–68, 2015.
- [8] M. Kitsak *et al.* Edge-level Collective Influence: skin and body variants for ℓ -ball neighbourhoods. *Entropy*, 26(3):248, 2024. <https://www.mdpi.com/1099-4300/26/3/248>.
- [9] Authors of MDPI *Entropy* 26(4) 315. Link-Local Betweenness Centrality and its mapping entropy extension via the first-order central domain. *Entropy*, 26(4):315, 2024. <https://www.mdpi.com/1099-4300/26/4/315>.
- [10] M. Toyoda *et al.* Topological analysis of the Japanese high-voltage transmission network. *IEEJ Trans. Power and Energy*, 128(3), 2008 (in Japanese).